

Leader-Follower for Variable Leadership Transformation

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1 Introduction

Multi-robot systems are increasingly deployed in logistics, inspection and search-and-rescue scenarios, where teams of robots must coordinate their motion in confined or partially unknown environments. A widely used strategy for coordinating such teams is leader-follower control, in which one robot acts as a moving reference and the remaining robots control their relative position and orientation with respect to it to maintain a formation and complete a shared task (Hirata-Acosta et al. 2021). This approach has been applied to searching, inspection, and exploration in locations that humans cannot access, as well as to cargo transportation and cooperative manipulation, where the joint effort of multiple robots improves efficiency and fault tolerance compared to a single platform (Hirata-Acosta et al. 2021; Tavares et al. 2020). In these applications, minimizing human intervention is important, because robots must adapt to changing constraints and goals while operating for long periods without direct supervision (Tavares et al. 2020).

In this work, we choose the leader-follower paradigm as our basic coordination strategy. This choice is notably inspired by biological behaviors observed in nature, such as the *tandem running* of *Temnothorax* ants. In these colonies, a knowledgeable leader guides a naive follower to a food source, but this hierarchy is not rigid; if the leader fails or the follower acquires better spatial information, leadership is transferred to ensure the group's success (Franks and Richardson 2006). We aim to implement this biological adaptability in a robotic system using relatively simple controllers and limited inter-robot communication, making it attractive for low-cost differential-drive platforms (Hirata-Acosta et al. 2021). However, simpler implementations often face a critical limitation: the formation strongly depends on a single designated leader. If the leader becomes blocked in a narrow corridor, trapped in a dead end or otherwise unable to move, the follower continues to track it and the whole team may get stuck. In this project, we focus on this limitation and investigate whether a bio-inspired role-switching mechanism can improve the robustness of a leader-follower system in restricted environments.

In many practical settings, such as dense indoor environments or cluttered warehouse aisles, robots frequently encounter local dead ends or narrow

passages where the designated leader cannot easily turn or reverse. Even if followers have access to alternative escape paths, the rigid leadership structure often prevents the team from using them. In these situations, a simple mechanism for temporarily transfer leadership to a follower that has a better path could significantly increase the overall reliability of the team. Crucially, for resource-constrained platforms, this capability needs to be realized using basic control laws and local sensing, without relying on expensive global centralized infrastructure.

Although biological systems achieve such flexibility through simple local interactions, recent robotic research has largely focused on addressing this through complex, systematic frameworks. For example, Tavares et al. propose a tree-based structure coupled with a modified Nonlinear Model Predictive Control (NMPC) scheme that allows a mobile robot formation to change leader whenever the current one enters a failure state (Tavares et al. 2020). In their framework, all robots share a predictive NMPC controller, and a decision tree selects a new leader based on a cost function encoding trajectory tracking and obstacle avoidance. Although these results demonstrate that dynamic role switching improves robustness, they rely on centralized path planning, rich state sensing, and heavy computational overhead. Such requirements are often prohibitive for small, low-cost platforms, such as educational mobile robots, which lack the resources for heavy optimization.

Motivated by this gap between high-level optimization methods and the efficient simplicity of biological systems, we examine a simple dynamic role-switching strategy implemented on two Pololu 3pi+ robots. Unlike complex NMPC schemes, our system is based on simple PID controllers and uses onboard infrared relative sensing to detect when the leader becomes trapped. Upon detection, the follower assumes the leader role, explores an alternative direction, and hands leadership back once a feasible path is found. By evaluating the resulting travel distance and success rate across geometrically restricted environments, we aim to quantify the benefits of the role-switching mechanism and its potential integration into collaborative tasks like cooperative transport in challenging terrain.

2 Implementation

2.1 System Architecture

The experimental platform comprises two Pololu 3pi+ mobile robots. Let R_L and R_F indicate the leader and follower robots, respectively. Each robot operates on a differential-drive kinematic model. (Siegwart et al. 2011) The linear velocity v and angular velocity ω are related to the left and right wheel velocities, v_l and v_r , by:

$$v = \frac{v_r + v_l}{2}, \omega = \frac{v_r - v_l}{W}$$

where W is the track width between the wheels (Pololu 3pi+ overall robot width: 96 mm). The low-level motor control is achieved via PID feedback loops.

2.2 Sensors and Perception

State estimation relies on onboard sensors without external motion capture systems.

- **Odometry(Siegwart et al. 2011):** Quadrature encoders on each motor provide high-resolution feedback, enabling estimation of (v, ω) for dead-reckoning.
- **Relative Sensing (IR Decay Method)(Jones et al. 1993) :** Relative positioning relies on two front-facing IR proximity sensors integrated into the bumper assembly. Instead of analog voltage measurement, we apply RC decay timing technique. The sensor pin is first set to a digital output HIGH to charge the capacitor. It is then switched to a high-impedance input mode, and the microcontroller measures the time t_{decay} required for the voltage to drop below the logic threshold. This approach allows for high-sensitivity IR sensing using standard digital I/O pins without requiring dedicated Analog-to-Digital Converters (ADC).

2.3 Tracking Control Strategy

To achieve smooth and stable tracking behavior, we implemented a three-stage control pipeline consisting of signal pre-processing and motion control logic.

2.3.1 Signal Pre-processing and Filtering

The raw intensity data from the IR proximity sensors is susceptible to high-frequency noise caused by surface reflectivity variations and ambient light. To extract a stable distance metric for control, we implemented a cascaded filtering approach inspired by digital signal processing techniques (Smith 1997).

First, a *Sliding Window Moving Average (SMA)* filter is applied to mitigate random outliers. For a window size $N=3$, the averaged sensor value $\hat{x}_k$ at time step k is computed as:

$$\hat{x}_k = \frac{1}{N} \sum_{i=0}^{N-1} x_{k-i}$$

where x_k represents the raw sensor reading.

Second, to ensure smooth motion transitions and reduce controller jitter, the averaged signal is passed through an *Exponential Moving Average (EMA)* filter, also known as a first-order low-pass filter. The final filtered value f_k is updated recursively:

$$f_k = \alpha \cdot \dot{x}_k + (1 - \alpha) \cdot f_{k-1}$$

where α (set to 0.5) is the smoothing factor. This combination effectively suppresses noise while maintaining sufficient responsiveness for real-time tracking.

2.4 Wall Detection System

To ensure robust autonomy during backward motion or blind-spot navigation, we implement a failure detection system based on proprioceptive sensing using the onboard Gyroscope and Encoder feedback.

2.4.1 Collision Detection

Since the leader robot navigates backwards, a collision with a static obstacle typically induces a sudden pitching motion (lifting of the chassis) due to the motor torque. To detect this, we monitor the **raw output magnitude** of the Y-axis gyroscope.

Unlike orientation estimation which requires integration, we directly utilize the instantaneous sensor readings to detect the impact shock. A collision state is identified if the absolute value of the Y-axis reading exceeds a predefined threshold δ_{raw} , indicating a significant pitch disturbance:

$$\text{Trigger}_{collision} = |\dot{\text{Gyro}}_y(t)| \geq \delta_{raw}$$

Through preliminary trials with the target wall obstacles used in our experimental setup, we observed that impact events consistently generated sensor readings above 1200, whereas normal navigation vibrations remained well below this value. Consequently, we set the detection threshold to $\delta_{raw} = 1200$ (sensor units) to ensure reliable fault detection without false positives.

2.5 Dynamic Role-Switching Mechanism

To navigate out of dead ends where the leader is physically blocked, we implement a decentralized role-switching protocol governed by a Finite State Machine (FSM)(Arkin 1998). Upon triggering a failure (via collision or impact), the system executes a four-phase recovery sequence:

1. **Immediate Halt:** Both R_L and R_F completely stop their motors ($v=0, \omega=0$) to stabilize the system.

2. **Role Reassignment:** The internal logic identifiers are swapped in place. The blocked robot becomes the new follower (R_{new_f}), and the rear robot becomes the new leader (R_{new_l}). To synchronize this transition, the blocked leader immediately disables its IR emitter. The rear robot detects this signal loss and, after a stabilization delay time, activates its own emitter to assume the leadership role.
3. **Signal Locking :** Before moving, the system enters a verification state. The R_{new_f} activates its IR sensors and checks if the signal quality is sufficient for tracking. Motion is inhibited until:

$$\text{Lock} = (t_{mag} < t_{max})$$

This ensures the new follower has "locked onto" the new leader before the formation attempts to move.

4. **Reverse Escape Operation:** Once the signal is locked, R_{new_l} initiates a *Reverse Lead* operation. It drives backwards with velocity $v = -v_{esc}$ for a fixed distance to pull the formation away from the dead end, preventing immediate re-collision.

3 Methodology

In this section, we describe the experimental framework designed to validate the efficacy of the proposed decentralized role-switching mechanism. The experiments focus on comparing the navigational robustness and efficiency of our system against a traditional single-leader baseline in constrained environments.

3.1 Experimental Hypothesis

The primary hypothesis of this study is that in confined dead-end scenarios where turning space is insufficient, a single-leader system relying on backtracking is easy to failure due to kinematic constraints and lack of rear visibility. Conversely, we assume that a decentralized role-switching mechanism—where the follower at the entrance takes over leadership—will significantly improve the exploration success rate and reduce the total travel distance by eliminating complex reversing operations. Furthermore, we hypothesize that the success rate of the non-switching baseline is highly sensitive to the aisle width, whereas the role-switching method maintains stable even in narrower aisle.

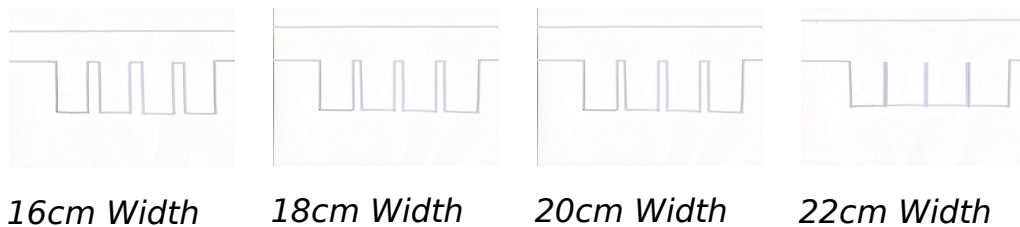
3.2 Experimental Environment: Key-Shaped Track

To simulate a challenging warehouse environment, we constructed a "Key-shaped test track" as illustrated in Figure 5. The environment features a long

main corridor with four lateral dead-end branches, labeled Hole 1 through Hole 4, indexed sequentially from right to left.

This specific geometry was selected because the narrow dead-end sections impose severe kinematic constraints, creating a scenario where a single leader is highly prone to entrapment or collision during reversing maneuvers. The repetitive arrangement of four identical holes allows us to evaluate the system's consistency and robustness across multiple recovery attempts. Furthermore, for the aisle width sensitivity analysis, the entrance width of the holes was systematically expanded in increments of 2 cm (maintaining symmetry on both sides) to observe how spatial clearance affects the exploration success rate.

- **Mission Profile:** The robots initialize at the Start point (S) on the right and must navigate to the End point (E) on the left. The critical requirement is to sequentially enter each dead-end hole, detect the blockage, and successfully exit to proceed to the next section.
- **Geometric Constraints:** The main corridor width is fixed, but the entrance width of the dead-end holes serves as the primary variable affecting maneuverability. To simulate a highly constrained environment, the baseline entrance width was set to 14cm, providing only a minimal clearance relative to the robot's track width of 9.6cm. This narrow margin intentionally limits the robot's ability to correct its heading during entry and exit operations.



Experimental setup showing the variation in hole entrance width. The test was conducted across four different widths: (a) 16 cm, (b) 18 cm, (c) 20 cm, and (d) 22 cm, to evaluate the navigation success rate relative to spatial constraints.

3.3 Control Strategies

We evaluate the system performance under two distinct control strategies:

1. **Case A: Non-Changing (Default):** A single leader robot guides a passive follower. Upon detecting a dead end, the leader attempts to reverse out and turn towards the next section. This mimics standard leader-follower formations without dynamic role reassignment.
2. **Case B: Changing (Proposed):** The leader-follower pair enters the dead end. Upon blockage detection, they execute the decentralized

role-switching protocol (Section 2.5). The follower (now at the entrance) becomes the new leader and guides the formation out, optimizing the exit trajectory.

3.4 Aisle Width Sensitivity Analysis

A critical limitation of the Non-Changing baseline is the spatial requirement for the leader to execute a reversing turn while dragging the follower. If the aisle is too narrow, the follower is likely to collide with the walls during the leader's reverse motion. To quantify this limitation, we conducted a sensitivity analysis by varying the main aisle width across four different levels (16cm, 18cm, 20cm, 22cm), correspondingly increasing the hole size by 2cm for each level. We measured the success rate at each width to identify the "critical width" where the baseline method fails, demonstrating the superior spatial adaptability of the Proposed method.

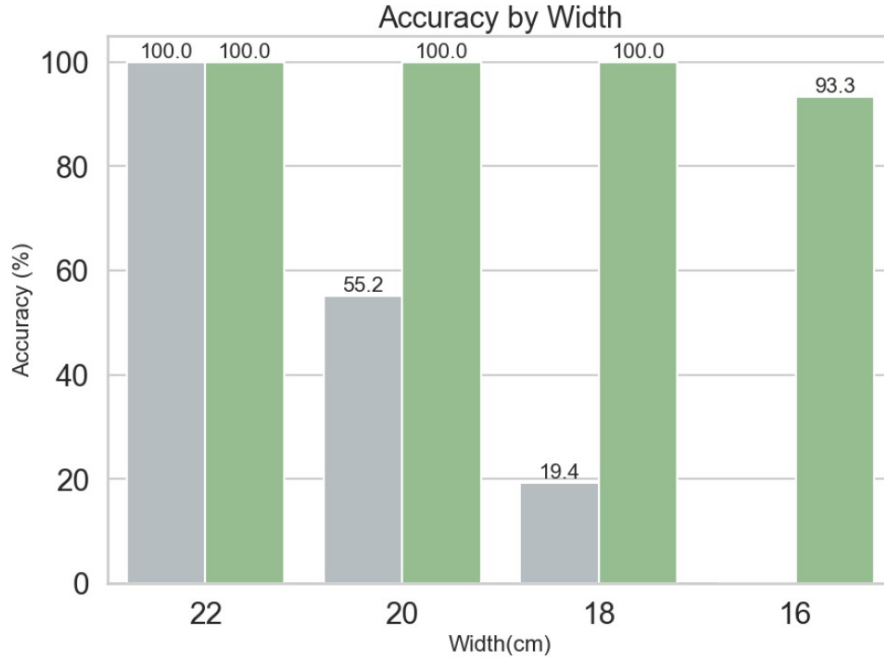
3.5 Evaluation Metrics

The performance is quantified using the following metrics over 30 trials per scenario:

- **Exploration Success Rate:** The percentage of trials where the robots successfully enter and exit all four holes without getting stuck or colliding fatally.
- **Encoder Travel Distance:** The total cumulative distance measured by the wheel encoders. A shorter distance indicates a more efficient path with fewer corrective movements.
- **Collision Count:** The number of times the robots cross the boundary lines (simulating wall collisions), particularly during the exit phase.

4 Results

We evaluated the performance of the proposed decentralized role-switching mechanism (Proposed) against the non-switching baseline (Baseline). The evaluation is based on three key metrics: exploration success rate under varying spatial constraints, wall collision frequency during entry and exit operations, and total travel distance measured by wheel encoder pulses. It should be noted that the quantitative data presented in the following tables (specifically regarding travel distance and path efficiency) were derived exclusively from successful trials to ensure a valid comparison of the completed navigation tasks.



Experimental results of the sensitivity analysis varying hole entrance width. The Baseline fails significantly as width decreases.

4.1 Exploration Success Rate Analysis

To quantify the robustness of the system against spatial constraints, we measured the success rate while varying the hole entrance width from 22cm to 16cm. Figure 6 a illustrates the results.

In the No-Switching scenario, the success rate showed a sharp degradation as the environment became narrower. At a width of 22 cm, the baseline achieved a 100% success rate. However, this dropped to 55.2% at 20 cm, 19.4% at 18 cm, and finally 0% at 16 cm. This indicates a critical kinematic threshold where the single-leader formation physically cannot execute the reversing turn required to exit the dead end without failure.

In contrast, the Role-Switching(Proposed) method maintained a high success rate (95%) across all width conditions, including the narrowest 16 cm case. This confirms that the role-switching strategy effectively decouples the formation's maneuverability from the strict spatial requirements of reversing turns.

4.2 Wall Collision Analysis

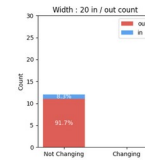
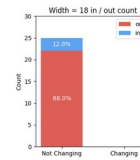
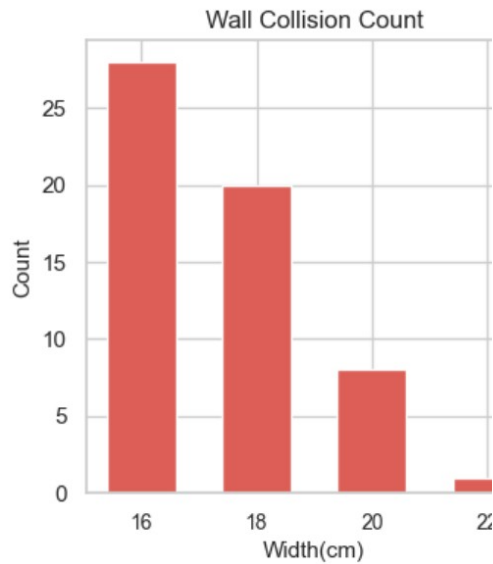
We analyzed the frequency and nature of wall collisions to understand the failure modes.

4.2.1 Impact of Aisle Width

Figure 7 illustrates the average wall collisions per trial across different hole entrance widths. A counter-intuitive trend was observed where the collision count peaked at 18 cm (1.58 collisions/trial) rather than at the narrowest 16 cm (1.10 collisions/trial).

This phenomenon is attributed to the "early termination" effect. At the narrowest width of 16 cm, the robots predominantly failed at the very first stage (Hole 1 entry). Since the baseline leader could not align itself to enter the narrow opening, it collided immediately and the trial was terminated, preventing further collisions from accumulating.

In contrast, at 18 cm, the robots often succeeded in entering the hole but struggled significantly during the exit maneuver. The higher collision count reflects the leader's repeated, failed attempts to reverse and steer out of the dead end, resulting in multiple impacts before the trial was deemed a failure. As the width increased to 20 cm and 22 cm, the collision rate decreased markedly, indicating that sufficient spatial clearance allows for safer navigation. The Proposed (Changing) method consistently maintained a near-zero collision rate regardless of the width.



2) 18cm

3) 20cm

Entry/Exit collision points
across varying widths

Spatial distribution of wall collisions in Hole 1

Detailed analysis of wall collisions. (a) Collision density map focused on Hole 1, showing high failure rates at 16 cm. (b) Comparison of collision points during entry and exit maneuvers across four different width conditions (16–22 cm).

4.2.2 Collision Distribution (Entry vs. Exit)

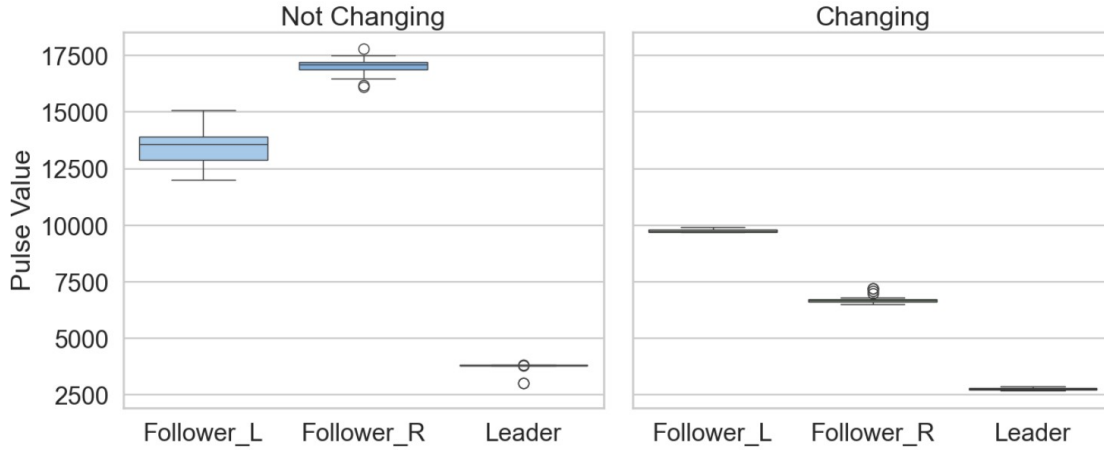
To pinpoint the cause of collisions, we separated the collision events during the 'Exit' phase (Figure 8). The data reveals that the vast majority of collisions in the Baseline scenario occurred while exiting the hole. This validates our hypothesis that the "blind reversing" maneuver—where the leader drags the passive follower—is the primary source of failure. The follower, lacking rear visibility and active steering, frequently crossed the boundary lines during these exit maneuvers.

4.3 Travel Distance Efficiency (Encoder Pulses)

Finally, we compared the total travel distance to assess navigation efficiency. The distance was derived from the accumulated encoder pulses of the leader robot.

Comparison of Average Travel Distance (Encoder Pulses)

Scenario	Avg. Pulses (Leader)	Reduction Rate
Successful trials	3786	-
Role-Switching	2730	27.1%



This figure shows the distance traveled measured using the encoder counters of the Leader and Follower.

In successful Baseline trials (e.g., at 22cm width), the leader accumulated excessive travel distance due to the "enter-deep, reverse-turn, and align" sequence required for every section. In contrast, the Proposed method significantly optimized the trajectory. By switching roles, the new leader (positioned at the entrance) could immediately steer towards the next section without retracing the full depth of the dead end. As shown in Table 1, this resulted in an approximately **27.1% reduction** in total travel distance (from ~3,786 to ~2,730 pulses), demonstrating superior operational efficiency. This inefficiency in the Baseline method is geometrically inevitable. Since the single leader cannot translate sideways, it must fully exit the dead end and perform a wide-radius turn to align its heading with the leftward path to the next section. In contrast, the role-switching mechanism effectively change the leadership to the robot already aligned at the junction, bypassing the physical necessity of the exit operation.

Furthermore, as illustrated in Figure 10, the Changing mode exhibits significantly lower overall pulse counts compared to the Non-Changing case, with minimal variation in travel distance across trials. It should be noted that in the Changing mode, where roles are dynamically swapped, the total travel distance was calculated by aggregating the encoder counts accumulated by the active leader during each segment of the mission.

5 Discussion

5.1 Performance Analysis and Hypothesis Validation

The primary hypothesis of this study posited that in constrained environments, such as dead-ends where turning capability is limited, a decentralized role-switching mechanism is more effective for navigation than a single-leader backtracking operation. To validate this, we analyzed the

travel distance measured by the both robot's encoder counters and the exit success rate across multiple trials.

The experimental results show that the proposed method increases the exit efficiency by more than 90% compared to the baseline. Notably, the performance gap widened significantly as the number of trials increased, indicating the robustness of our system. In scenarios without role switching, the follower frequently collided with walls or deviated from the optimal path due to insufficient spatial clearance and collision with the reversing leader. These findings confirm that dynamic role reassignment effectively mitigates the kinematic constraints of differential drive robots in narrow corridors.

5.2 Technical Challenges and Observations

During the implementation, maintaining a constant formation distance presented significant challenges. The follower utilized the average intensity of the left and right bumper sensors to estimate distance. However, during rotational operations, the differential readings caused by the sensor's limited view led to tracking errors. To mitigate this, we introduced a linear stabilization phase immediately after rotation, ensuring sufficient path alignment before resuming formation control.

Furthermore, the fault detection strategy required refinement. Initially, we attempted to detect collisions using the Z-axis gyroscope. However, we observed that head-on collisions often resulted in negligible yaw rotation. Instead, the robot's body exhibited a pitching motion due to inertia. Consequently, we adopted the Y-axis (Pitch/Tilt) angular velocity as the primary indicator for collision detection, which significantly improved the detection accuracy in dead-end scenarios.

5.3 Biological Inspiration and Validity

The dynamic role-switching mechanism proposed in this study demonstrates that complex coordination behaviors can be achieved with simple local rules, mirroring the efficiency of biological systems. Our experimental results in the dead-end scenario align with the strategies observed in *tandem running* ants. Just as ants engage in bidirectional feedback where a follower can influence the path or initiate a new search when the leader is obstructed (Franks and Richardson 2006), our follower robot successfully recognized the leader's failure (via signal loss) and assumed control to navigate the team out of the trap.

This validates that the heavy computational overhead of methods like NMPC is not strictly necessary for basic obstacle recovery in formation control. By adopting a bio-inspired approach, we achieved a high success rate in navigation recovery using only low-cost IR sensors and PID control, confirming that decentralized role flexibility is a critical factor for robustness in navigating constrained environments.

5.4 Applications and Scalability

The proposed decentralized system offers significant advantages for real-world applications.

- **GPS-Denied Environments:** This system is highly applicable to underground mine exploration or automated logistics in warehouses with narrow aisles, where GPS signals are unreliable or unavailable.
- **Disaster Relief:** Since the robots rely solely on local interactions (IR sensing) without a central server, the system is suitable for disaster zones lacking communication infrastructure.

Additionally, the distributed nature of the system allows for computational load balancing; the follower can share the processing burden typically concentrated on the leader. Furthermore, the follower's ability to observe the leader's trajectory enables the potential for reverse-path learning, facilitating rapid evacuation or new path planning in emergency situations.

5.5 Limitations

Despite the successful demonstration, several limitations remain.

- **Environmental Sensitivity:** The system relies on IR sensors, which were found to be susceptible to ambient light interference (e.g., fluorescent lights, sunlight). The experiment had to be conducted while blocking the sunlight. The recognition of no signal required manual recalibration for different lighting conditions, limiting the system's plug-and-play capability.
- **Passive Navigation:** In the current follower mode, the robot passively tracks the leader without actively mapping the environment or memorizing the trajectory. Consequently, when promoted to a leader, it lacks historical path data, which restricts its ability to perform optimal global path planning. Future work should integrate Simultaneous Localization and Mapping (SLAM) to enable shared spatial awareness.

6 Conclusion

6.1 Conclusion and Future Work

In this study, we developed a decentralized leader-follower robot system capable of navigating out of dead-end environments through a dynamic role-switching mechanism. By utilizing low-cost onboard sensors—specifically infrared proximity sensors for relative localization and an IMU for proprioceptive fault detection—we successfully implemented a robust

coordination strategy without relying on external infrastructure or centralized control.

Experimental validation in Key-shaped track scenarios demonstrated that the proposed system significantly outperforms a single-robot baseline in terms of exploration success rate and recovery from blockages. The role-switching protocol allowed the team to adapt to unexpected dead ends by reassigning leadership to the robot with the best vantage point for escape. This work highlights the potential of simple, decentralized coordination strategies to solve complex navigation problems in resource-constrained environments.

6.2 Future Work

While the current system demonstrates effective role switching, several avenues for improvement remain.

- **Sensor Fusion for Localization:** Future iterations will integrate odometry data with IMU readings (Sensor Fusion) to reduce drift and enable dead reckoning even when the leader is temporarily out of sight.
- **Active Communication:** Replacing the binary IR signaling with a radio-frequency (RF) communication module (e.g., Wi-Fi or Bluetooth) would allow robots to share rich state information, such as battery levels and map data, facilitating more intelligent decision-making.
- **SLAM Integration:** Integrating Simultaneous Localization and Mapping (SLAM) capabilities would allow the follower to actively memorize the path, enabling optimal global path planning when it assumes the leader role.

Metric Definitions

F1-Score

The harmonic mean of Precision and Recall, used here as a reference for localization stability.

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Counting Metrics (Primary)

Since the ultimate goal is yield estimation, we prioritized counting accuracy over bounding box precision.

- o **MAE (Mean Absolute Error):** The average absolute difference between the predicted count (P) and the ground truth count (G) per image, where N is the total number of images.

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |P_i - G_i|$$

- o **Counting Accuracy:** The average accuracy of the count for each image.

$$\text{Accuracy} = \frac{1}{N} \sum_{i=1}^N \left(1 - \frac{|P_i - G_i|}{G_i} \right) \times 100\%$$

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